Abel Lynerd

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Summary

I am an Echols Scholar studying mathematics in the University of Virginia's College of Arts and Sciences. My primary interests lie in mathematical research (specifically in operator theory and combinatorics). I aspire to pursue graduate studies, with the ultimate goal of becoming a math professor.

Education

2025 - present	B.A. Mathematics candidate at the University of Virginia	(GPA: 3.797/4.00)
2021 - 2025	High School Diploma at Warwick High	(GPA: 4.61/4.00)
2023 - 2025	International Baccalaureate Diploma Programme	(Diploma Received)

Honors and Awards

Spring 2027 Participant, Budapest Semesters in Mathematics — October 2026

Joint Mathematics Meetings Presenter, American Mathematical Society — October 2026

Summer Opportunity for Academic Research Virtual Program, Mathematics — May - Jul 2026
Laney Graduate School of Emory University

Echols Scholar, UVA College of Arts & Sciences — Aug 2026

Beard/Bennett Memorial Scholar, BayPort Credit Union — Apr 2025

H. Kruger Kaprielian Memorial Scholar, UVA Alumni Association — June 2025

Relevant Coursework

Fall 2026	MATH 4770 – General Topology	(Grade: In Progress)
Fall 2026	MATH 4310 – Introduction to Real Analysis	(Grade: In Progress)
Fall 2026	MATH 3354 – Survey of Algebra	(Grade: In Progress)
Spring 2026	MATH 3310 – Basic Real Analysis	(Grade: A)

Projects

Emory-LGS SOARVS REU — May - Aug 2026

In the summer after my first year, I was selected to participate in the Summer Opportunity for Academic Research Virtual Scholarly Program with the **Laney Graduate School of Emory University**. I attended weekly seminars and worked on various projects to develop my skills as a researcher, while also exploring special topics under Emory graduate students. For one project, I created and presented a literature review for a hypothetical paper on coding theory, drawing from personal study and interviews I'd held with faculty at various universities.

Independent research in graph theory and combinatorics — Mar 2026 - Present

Throughout the spring and summer of 2026, I have conducted research in graph theory and combinatorics, primarily in **Ramsey Theory and edge-coloring problems**. I undertook this project after interviewing **Professor Alexander Halperin** from Salisbury University, who introduced me to the n-tuple graph operation from one of his papers. I have found results on this operation's regularity and invertibility, and my work was **selected for presentation at the 2027 Joint Mathematics Meetings**.

Directed Reading Program on knot theory Jan - May 2026

In my second semester I studyied in UVA's Directed Reading Program under Adam Friedman-Brown, a graduate student studying topology. We read **Colin Adams's *The Knot Book***, meeting weekly to discuss. At the end of this semester I gave a talk on knot theory to the Math Department.

Math Experimental Lab on discrete dynamical systems Sep - Dec 2025

In my first semester I participated in UVA's Math Experimental Lab on discrete dynamical systems under **Dr. Oliver Wang**. In our weekly meetings we studied point behaviors for iterated functions in real and complex variables, and at the end of the semester we presented to the department. I received a grade of A for my work.

Academic Involvement

MATH 1191, Embedded Tutor Sep 2025 - present

I am a part of the team running MATH 1191: Survey of Calculus, assisting students in strengthening their mathematical background. My responsibilities include working with students in the classroom, as well as hosting office hours.

UVA Advancement, Office of Engagement Intern Mar 2026 - present

Responsible for creating a more efficient workflow for shipping supply packages to alumni clubs around the globe. This involves writing programs and macros to streamline the registration and labeling process, as well as extensive work in Microsoft Excel to oversee financial and progress reports.

UVA Undergraduate Math Club and Putnam Team, Social Chair Sep 2025 - present

Responsible for organizing social events and fostering undergraduate engagement in UVA's mathematical community.

Prof. Richard Schwartz's Virginia Mathematics Lectures, University of Virginia Mar 2026

I attended Brown Professor Richard Schwartz's lectures on the most efficient paper moebius band and the optimal origami torus.

Prof. Ravi Vakil's Virginia Mathematics Lectures, University of Virginia Sep 2025

I attended Stanford Professor and AMS President Ravi Vakil's lecture on the Mathematics of Doodling.

Relevant Skills

- Proficient in LaTeX
- Conversational in French